Project Report

IITM BS Data Science – Modern Application Development II

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Project Title: Vehicle Parking App

Description

The project aims to develop a full-stack vehicle parking management system for 4-wheeler vehicles. The app supports both admin and user functionalities, and provides features such as lot creation, spot allocation, reservation tracking, and CSV report generation.

Technologies Used

- **Flask** – For building backend RESTful APIs
- **Vue.js** – For building the frontend user interface
- **SQLite** – Lightweight database for data persistence
- **Redis** – For Celery task queue broker
- **Celery** – For scheduling periodic email reports
- **Chart.js** – To display reservation statistics
- **Flask Extensions:**
  - `flask-restful` – Structuring API endpoints
  - `flask-jwt-extended` – User authentication and token handling
  - `flask-caching` – Optimizing API responses

DB Schema Design

The database uses the following main tables:

- **User** – Stores user info with role (‘is_admin‘) and credentials.
- **ParkingLot** – Holds lot name, ID, and number of spots.
- **ParkingSpot** – Contains individual spots linked to a lot.
- **Reservation** – Manages reservation records, timestamps, and cost.

All reservations are soft-deleted (marked inactive) when ended. This preserves history for reporting and avoids foreign key issues if lots are removed.

API Design

The app exposes RESTful endpoints for:

- `/register`, `/login` – User authentication
- `/lots`, `/lots/<lot_id>` – CRUD for parking lots
- `/spots` – Fetch all spots
- `/reservations` – Auto-assign and create reservations
- `/reservations/<id>` – End reservations
- `/generate_report/<user_id>` – Generate and email monthly CSV
- `/reservations/<id>/csv` – Download reservation CSV

  All APIs return JSON and use JWT tokens for authentication.

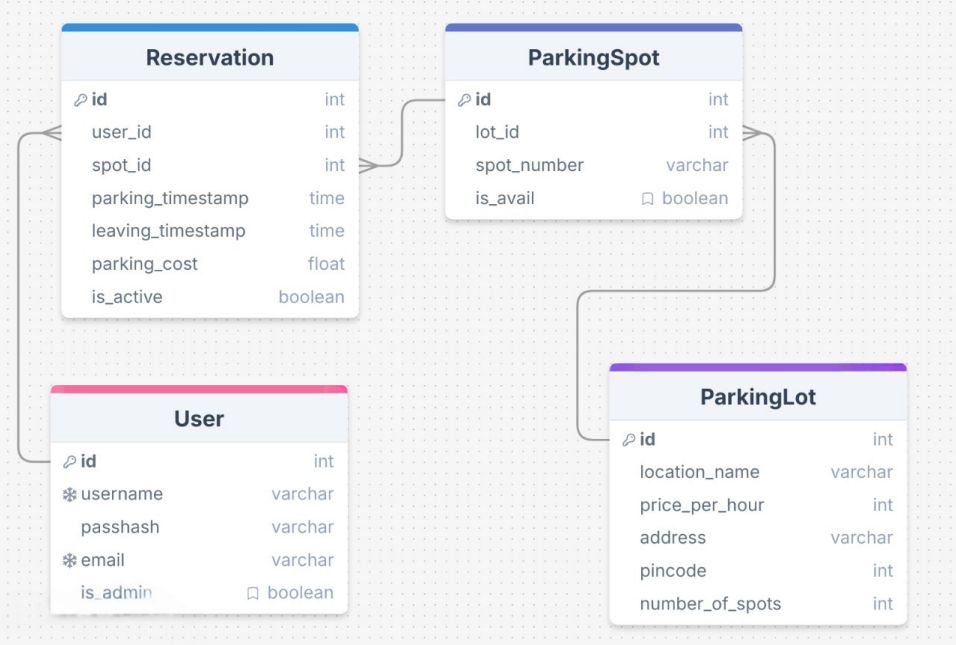
Architecture and Features

The backend (Flask) is organized with separate modules for models, resources (API), and Celery tasks. Frontend (Vue.js) uses components to manage dashboard views and user actions.

Key Features:

- Admin dashboard to manage lots and monitor spot availability.
- Auto-spot assignment to ensure fairness.
- Soft deletion of reservations to maintain historical records.
- Daily and monthly email reminders via Celery.
- CSV export of reservation data.
- Interactive graphs using Chart.js.

ER Diagram



Video

The project presentation video can be viewed at the following link: `https://drive.google.com/file/d/1rMNTUXxPZFsfv1JLBerz3tTfGuedGncQ/view?usp=sharing`

AI/LLM Acknowledgement

I used ChatGPT to debug backend logic and for implementing some CSS styling enhancements.